

User: Projet VECM
Project: 1

```
1 . import excel "C:\Users\LENOVO\Downloads\Base de donnée VECM.xlsx", sheet("Feuill") firstrow clear
2 . destring, replace
   ANNEE already numeric; no replace
   PIB: all characters numeric; replaced as double
   ide: all characters numeric; replaced as double
   eib: all characters numeric; replaced as double
   ipc: all characters numeric; replaced as double
3 . tsset ANNEE
   time variable: ANNEE, 1985 to 2020
   delta: 1 unit
```

```
4 . summarize PIB ipc eib ide
```

Variable	Obs	Mean	Std. Dev.	Min	Max
PIB	36	6.34e+10	3.44e+10	1.50e+10	1.20e+11
ipc	36	84.55513	20.17558	45.14568	111.8517
eib	36	1.43e+10	7.69e+09	3.36e+09	2.80e+10
ide	36	2.45e+08	2.99e+08	-4487772	1.01e+09

```
5 . correlate PIB ipc eib ide
   (obs=36)
```

	PIB	ipc	eib	ide
PIB	1.0000			
ipc	0.9297	1.0000		
eib	0.9873	0.9278	1.0000	
ide	0.8789	0.7775	0.8943	1.0000

```
6 . dfuller PIB, trend regress lags(0)
```

Dickey-Fuller test for unit root Number of obs = 35

	Test Statistic	Interpolated Dickey-Fuller		
		1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-1.642	-4.288	-3.560	-3.216

MacKinnon approximate p-value for Z(t) = 0.7757

D.PIB	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
PIB						
L1.	-.148138	.0902326	-1.64	0.110	-.3319358	.0356597
_trend	4.86e+08	2.97e+08	1.64	0.112	-1.19e+08	1.09e+09
_cons	3.27e+09	1.65e+09	1.98	0.056	-9.47e+07	6.63e+09

```
7 . dfuller PIB, drift regress lags(0)
```

Dickey-Fuller test for unit root Number of obs = 35

	Test Statistic	1% Critical Value	Z(t) has t-distribution		10% Critical Value
			5% Critical Value		
Z(t)	-0.234	-2.445	-1.692		-1.308

p-value for Z(t) = 0.4081

11 . dfuller ipc, noconstant regress lags(0)

Dickey-Fuller test for unit root Number of obs = 35

		Interpolated Dickey-Fuller			
Test Statistic		1% Critical Value	5% Critical Value	10% Critical Value	
Z(t)		-2.644	-1.950	-1.604	
D.ipc	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
ipc L1.	.0198661	.002917	6.81	0.000	.013938 .0257942

12 . dfuller eib, trend regress lags(0)

Dickey-Fuller test for unit root Number of obs = 35

		Interpolated Dickey-Fuller			
Test Statistic		1% Critical Value	5% Critical Value	10% Critical Value	
Z(t)		-4.288	-3.560	-3.216	

MacKinnon approximate p-value for Z(t) = 0.5022

D.eib	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
eib L1.	-.2796419	.1283863	-2.18	0.037	-.5411562	-.0181276
_trend	1.94e+08	9.55e+07	2.03	0.051	-857616.6	3.88e+08
_cons	1.03e+09	5.39e+08	1.91	0.066	-7.08e+07	2.13e+09

13 . dfuller eib, drift regress lags(0)

Dickey-Fuller test for unit root Number of obs = 35

		Z(t) has t-distribution			
Test Statistic		1% Critical Value	5% Critical Value	10% Critical Value	
Z(t)		-2.445	-1.692	-1.308	

p-value for Z(t) = 0.2133

D.eib	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
eib L1.	-.0284812	.0353881	-0.80	0.427	-.1004788	.0435165
_cons	9.87e+08	5.64e+08	1.75	0.089	-1.60e+08	2.13e+09

14 . dfuller eib, noconstant regress lags(0)

Dickey-Fuller test for unit root Number of obs = 35

		Interpolated Dickey-Fuller			
Test Statistic		1% Critical Value	5% Critical Value	10% Critical Value	
Z(t)		-2.644	-1.950	-1.604	

D.eib	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
eib L1.	.0261436	.0171931	1.52	0.138	-.0087969	.0610841

```
15 . pperon eib
command pperon is unrecognized
r(199);
```

```
16 . pperron eib
```

```
Phillips-Perron test for unit root      Number of obs   =      35
                                         Newey-West lags =       3
```

Test Statistic	Interpolated Dickey-Fuller			
	1% Critical Value	5% Critical Value	10% Critical Value	
Z(rho)	-0.923	-17.880	-12.820	-10.400
Z(t)	-0.784	-3.682	-2.972	-2.618

MacKinnon approximate p-value for Z(t) = **0.8237**

```
17 . pperron ide
```

```
Phillips-Perron test for unit root      Number of obs   =      35
                                         Newey-West lags =       3
```

Test Statistic	Interpolated Dickey-Fuller			
	1% Critical Value	5% Critical Value	10% Critical Value	
Z(rho)	-4.683	-17.880	-12.820	-10.400
Z(t)	-1.566	-3.682	-2.972	-2.618

MacKinnon approximate p-value for Z(t) = **0.5008**

```
18 . pperron ipc
```

```
Phillips-Perron test for unit root      Number of obs   =      35
                                         Newey-West lags =       3
```

Test Statistic	Interpolated Dickey-Fuller			
	1% Critical Value	5% Critical Value	10% Critical Value	
Z(rho)	-1.163	-17.880	-12.820	-10.400
Z(t)	-3.117	-3.682	-2.972	-2.618

MacKinnon approximate p-value for Z(t) = **0.0253**

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19 . pperron PIB
```

```
Phillips-Perron test for unit root      Number of obs   =      35
                                         Newey-West lags =       3
```

Test Statistic	Interpolated Dickey-Fuller			
	1% Critical Value	5% Critical Value	10% Critical Value	
Z(rho)	-0.218	-17.880	-12.820	-10.400
Z(t)	-0.251	-3.682	-2.972	-2.618

MacKinnon approximate p-value for Z(t) = **0.9321**

20 . varsoc PIB ipc eib ide

Selection-order criteria
Sample: 1989 - 2020

Number of obs = 32

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-2280.82				1.2e+57	142.801	142.862	142.984
1	-2139.95	281.73	16	0.000	5.0e+53	134.997	135.3*	135.913*
2	-2130.76	18.381	16	0.302	8.1e+53	135.422	135.969	137.071
3	-2102.52	56.474	16	0.000	4.3e+53*	134.658*	135.447	137.039
4	-2088.77	27.505*	16	0.036	6.6e+53	134.798	135.83	137.913

Endogenous: PIB ipc eib ide
Exogenous: _cons

21 . vecrank PIB ipc eib ide, trend(trend) lags(3)

Johansen tests for cointegration

Trend: trend
Sample: 1988 - 2020
Number of obs = 33
Lags = 3

maximum rank	parms	LL	eigenvalue	trace statistic	5% critical value
0	40	-2202.8424	.	55.5446	54.64
1	47	-2184.3647	0.67367	18.5892*	34.55
2	52	-2177.3875	0.34483	4.6348	18.17
3	55	-2175.0701	0.13103	0.0000	3.74
4	56	-2175.0701	0.00000		

22 . vecstable

vecstable only works after vec
r(498);

23 . vec

vec results not found
r(301);

24 . vec PIB ide eib ipc

Vector error-correction model

Sample: 1987 - 2020
Number of obs = 34
AIC = 135.5122
Log likelihood = -2276.708
HQIC = 135.9256
Det(Sigma_ml) = 1.71e+53
SBIC = 136.7243

Equation	Parms	RMSE	R-sq	chi2	P>chi2
D_PIB	6	4.4e+09	0.4542	23.30172	0.0007
D_ide	6	1.6e+08	0.2730	10.51496	0.1046
D_eib	6	1.6e+09	0.2806	10.92087	0.0909
D_ipc	6	1.10733	0.7877	70.82397	0.0000

	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D_PIB						
_cel						
L1.	-.0930927	.0999347	-0.93	0.352	-.2889612	.1027757
PIB						
LD.	.2981297	.2615512	1.14	0.254	-.2145012	.8107606
ide						
LD.	17.57546	5.908867	2.97	0.003	5.994296	29.15663
eib						
LD.	-.6407329	.8412837	-0.76	0.446	-2.289619	1.008153
ipc						

LD.	8.25e+08	7.18e+08	1.15	0.250	-5.82e+08	2.23e+09
_cons	1.61e+08	1.62e+09	0.10	0.921	-3.02e+09	3.34e+09
D_ide						
_cel L1.	-.0051333	.0035511	-1.45	0.148	-.0120934	.0018268
PIB LD.	.0064537	.0092941	0.69	0.487	-.0117625	.0246699
ide LD.	-.2389373	.2099697	-1.14	0.255	-.6504703	.1725957
eib LD.	-.0583035	.0298947	-1.95	0.051	-.1168962	.0002891
ipc LD.	2.16e+07	2.55e+07	0.85	0.397	-2.84e+07	7.16e+07
_cons	-6603166	5.77e+07	-0.11	0.909	-1.20e+08	1.06e+08
D_eib						
_cel L1.	.0333355	.0352043	0.95	0.344	-.0356637	.1023346
PIB LD.	.1271388	.0921374	1.38	0.168	-.0534472	.3077248
ide LD.	3.619023	2.081534	1.74	0.082	-.4607085	7.698755
eib LD.	-.3047393	.2963615	-1.03	0.304	-.8855971	.2761186
ipc LD.	-4.61e+07	2.53e+08	-0.18	0.855	-5.42e+08	4.50e+08
_cons	4.50e+08	5.72e+08	0.79	0.432	-6.71e+08	1.57e+09
D_ipc						
_cel L1.	1.26e-11	2.50e-11	0.51	0.613	-3.63e-11	6.16e-11
PIB LD.	-1.12e-11	6.54e-11	-0.17	0.864	-1.39e-10	1.17e-10
ide LD.	1.25e-09	1.48e-09	0.85	0.396	-1.64e-09	4.15e-09
eib LD.	-8.64e-11	2.10e-10	-0.41	0.681	-4.99e-10	3.26e-10
ipc LD.	.4068105	.1794489	2.27	0.023	.0550971	.7585238
_cons	1.138212	.4057168	2.81	0.005	.3430212	1.933402

Cointegrating equations

Equation	Parms	chi2	P>chi2
_cel	3	405.4878	0.0000

Identification: beta is exactly identified

Johansen normalization restriction imposed						
beta	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
_ce1						
PIB	1	.	.	.	.	.
ide	47.36148	17.43408	2.72	0.007	13.19131	81.53164
eib	-6.811457	1.047892	-6.50	0.000	-8.865287	-4.757627
ipc	-1.35e-06	2.48e+08	-0.00	1.000	-4.87e+08	4.87e+08
_cons	2.16e+10	.	.	.	.	.

25 . vecstable

Eigenvalue stability condition

Eigenvalue	Modulus
1	1
1	1
1	1
.7264283	.726428
-.1572186 + .2817138i	.322615
-.1572186 - .2817138i	.322615
.2491087	.249109
-.06311173	.063112

The VECM specification imposes 3 unit moduli.

26 .